



Cambridge (CIE) IGCSE Computer Science



Your notes

Network Hardware

Contents

- * Network Interface Card (NIC)
- * MAC Addresses & IP Addresses
- * Router



Your notes

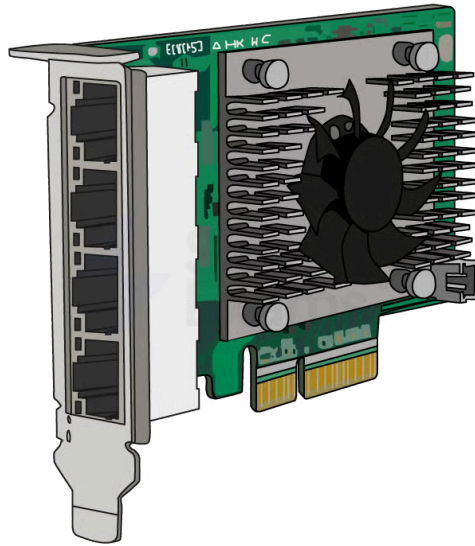
Network Interface Card (NIC)

Local Area Network Hardware: Network Interface Card (NIC)

- Network hardware is a selection of essential components that enable the **connectivity** and **communication** of devices within computer networks
- Networks use a variety of hardware to function, some of which include:
 - Router
 - Wireless access point (WAP)
 - Switch
 - Transmission media
 - Network interface card (NIC)
- The exam requires you to understand the purpose of the network interface card (NIC)

What is a network interface card (NIC)?

- The Network Interface Card (**NIC**) is required for a computer to connect to a network
- A NIC can be both wired and wireless and allows your computer to send and receive data over a network



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Your notes



Your notes

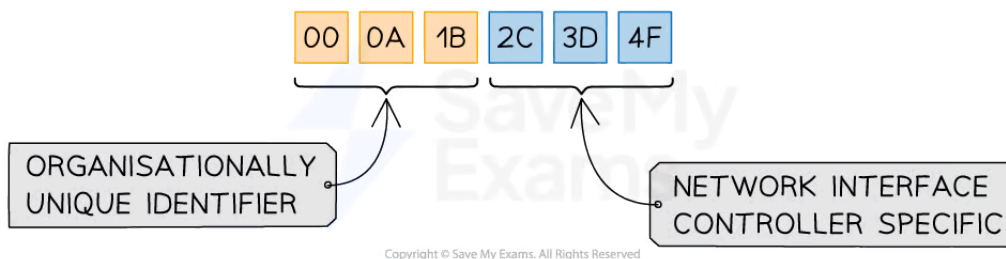
MAC Addresses & IP Addresses

- Devices on a network send and receive data, a device needs an address to ensure it sends data to the correct place
- There are two types of network address systems:
 - **MAC Address**
 - **IP Address**

MAC Addresses

What is a MAC address?

- A MAC (Media Access Control) address is a **unique identifier** given to devices which communicate over a local area network (**LAN**)
- A network interface card is given a MAC address at the point of manufacture
- MAC addresses are **static**, meaning they can never change
- MAC addresses make it possible for **switches** to efficiently **forward data** to the intended **recipient**
- Any device that contains a **Network Interface Card (NIC)** has a MAC address **assigned during manufacturing**
- A device connecting to a local network already has a MAC address, if it moves to a different network then the MAC address will stay the same



- A MAC address is represented as **12 hexadecimal digits** (48 bits), usually **grouped in pairs**
- The first three pairs are the **manufacturer ID number** (OUI) and the last three pairs are the **serial number** of the network interface card (**NIC**)
- There are enough unique MAC addresses for roughly **281 trillion devices**



Your notes

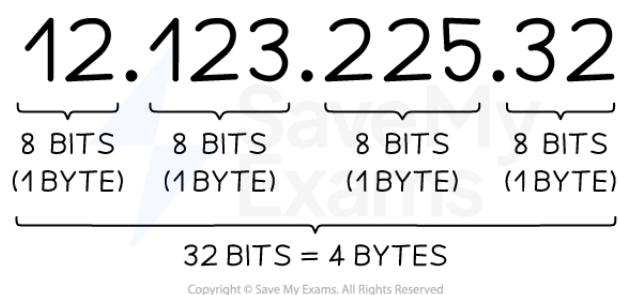
IP Addresses

What is an IP address?

- An IP (Internet Protocol) address is a **unique identifier** given to devices which communicate over the Internet (**WAN**)
- IP addresses can be **static**, meaning they stay the same or **dynamic**, meaning they can change
- IP addresses make it possible to **deliver data to the right device**
- A device connecting to a network will be given an IP address, if it moves to a different network then the **IP address will change**

IPv4

- Internet Protocol version 4 is represented as **4 blocks of denary numbers** between **0** and **255**, separated by **full stops**
- Each block is **one byte** (8 bits), each address is **4 bytes** (32 bits)



- IPv4 provides over **4 billion unique addresses** (2^{32}), however, with over **7 billion** people and **countless devices per person**, a solution was needed

IPv6

- Internet Protocol version 6 is represented as **8 blocks of 4 hexadecimal digits**, separated by **colons**
- Each block is **2 bytes** (16 bits), each address is **16 bytes** (128 bits)

2001:0DB8:3333:4444:5555:6666:7777:8888

16 BITS:16 BITS:16 BITS:16 BITS:16 BITS:16 BITS:16 BITS:16 BITS

128 BITS

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- IPv6 could provide over **one billion unique addresses** for **every person** on the planet (2^{128})



Worked Example

Computers in a network can be identified using both IP addresses and MAC addresses.

Describe two differences between IP addresses and MAC addresses [2]

Answer

- IP address is dynamic/can change // MAC address is static/cannot change
- IP address is used to communicate on a WAN/Internet // MAC address is used to communicate on a LAN



Your notes

Router

Router

What is a router?

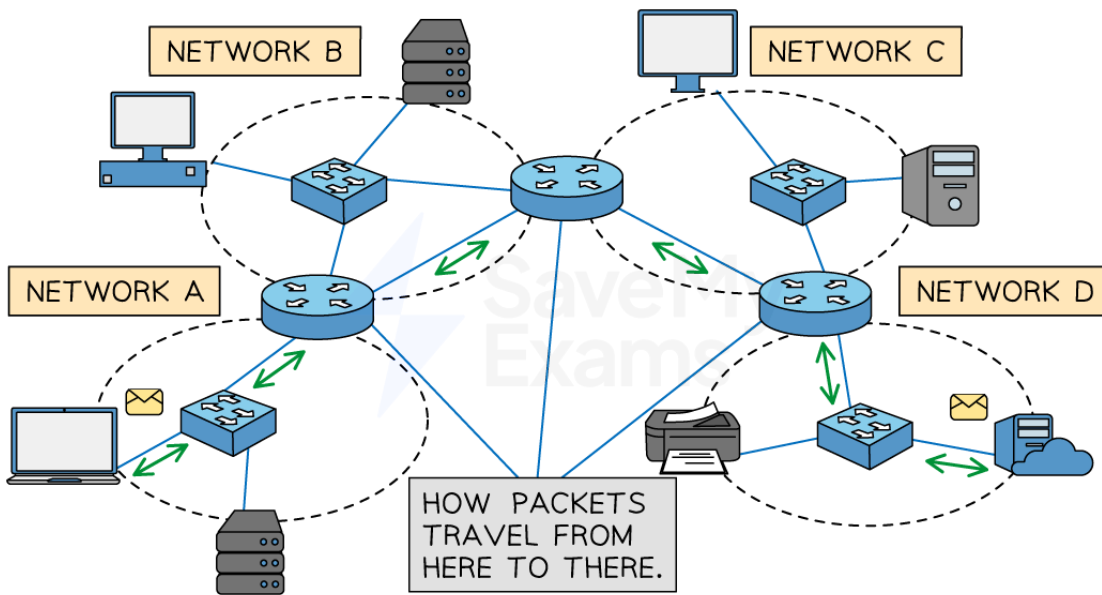
- The router is responsible for routing **data packets** between different networks
- An example of data the router can direct is, sending internet traffic to the correct destination/devices in your home network
- The router connects networks together, local area networks (LAN) to the wider internet which is a type of wide area network (WAN)
- The router can manage and prioritise data traffic, which can help to keep connections stable
- The router will assign IP addresses to the devices on the network

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Image of a router



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Image of how a router connects LANs to other networks



Worked Example

State 3 tasks carried out by a router [3]

To answer the question you must simply identify 3 tasks a router does

1 mark each to max 3 e.g.

- Send and Receive packets of data
- Connect a local network to the internet
- Assign IP addresses to nodes/devices
- Converts packets from one protocol to another